Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21311

F-411

B. C. A. (Third Sem.) Examination, 2022-23

(New Course)

Paper No. : BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) How many types of error? Explain any one with an example.

   (b) Describe in details numbers and their accuracy.

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2. (a) Find the absolute and relative error If $\pi=22/7$ is approximated as 3.14.

   (b) Find the real root of a equation $x^3-2x-5=0$ by the method of false position correct to three decimal places.

3. (a) Differentiate between Bisection method and Iteration method with an example.

   (b) Describe in details Newton Rapson method with an examples.

4. (a) The population of a town given below Estimate the population for the year 1925.

Year(x)	1891	1901	1911	1921	1931
Population (y)	48	68	83	95	103

   (b) Explain the rate of convergence of Iteractive methods with an example.

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5. (a) Using Newton's divided difference formula f(x) as a polynomial in x far the following data:

x	–4	–1	0	2	5
f(x)	1245	33	5	9	1335

(b) Explain the Gauss forward and back ward formula with an example.

6. (a) Use stirling's formula to find y_{35}, given $y_{20}=152$, $y_{30}=439$, $y_{40}=346$, and $y_{50}=243$.

(b) Differentiate between linear regression and line of regression.

7. (a) Use Weddle's rule to evaluate:

$$\int_0^{1/2} \frac{dx}{\sqrt{1+x^2}}$$

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using:

(i) Simpson's 1/3 rule taking h=1/4

(ii) Simposon's 3/8 rule taking h=1/6

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8. (a) Differentiate between Taylor's Method and Runge-Kutta Method with an example.

(b) Explain the following statistical computation.

(i) Frequency Chart.

(ii) Data fitting with cubic splines.

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